

Analysis for Cations

Qualitative analysis for cations is carried out in a slightly different way from that for anions. The common metallic actions are divided into a number of groups such that the metals in a particular group are precipitated by a particular reagent.

Group	Cations
I	Pb^{2+} , Hg_2^{2+} , Ag^+
II	Hg^{2+} , Pb^{2+} , Bi^{3+} , Cu^{2+} , Cd^{2+} , As^{3+} , Sn^{2+} , Sn^{4+} , Sb^{3+}
III	Al^{3+} , Cr^{3+} , Fe^{3+} , Mn^{2+}
IV	Co^{2+} , Zn^{2+} , Ni^{2+} , Mn^{2+}
V	Ba^{2+} , Sr^{2+} , Ca^{2+}
VI	Na^+ , K^+ , Mg^{2+} , NH_4^+

In general, the precipitating reagents take advantage of the varying solubility of metal chlorides, sulphides, hydroxides and carbonates in this scheme of separation. The only cation that is not metallic which is commonly encountered is the ammonium radical (NH_4^+) and this is detected in the original solution by boiling a small portion of it with sodium hydroxide and testing for NH_3 vapour as described earlier.

Test for NH_4^+

Treat 10 mg of mixture with 0.5 ml of NaOH and heat on a water bath. Check the evolved gas with filter paper moistened Nessler's reagent.

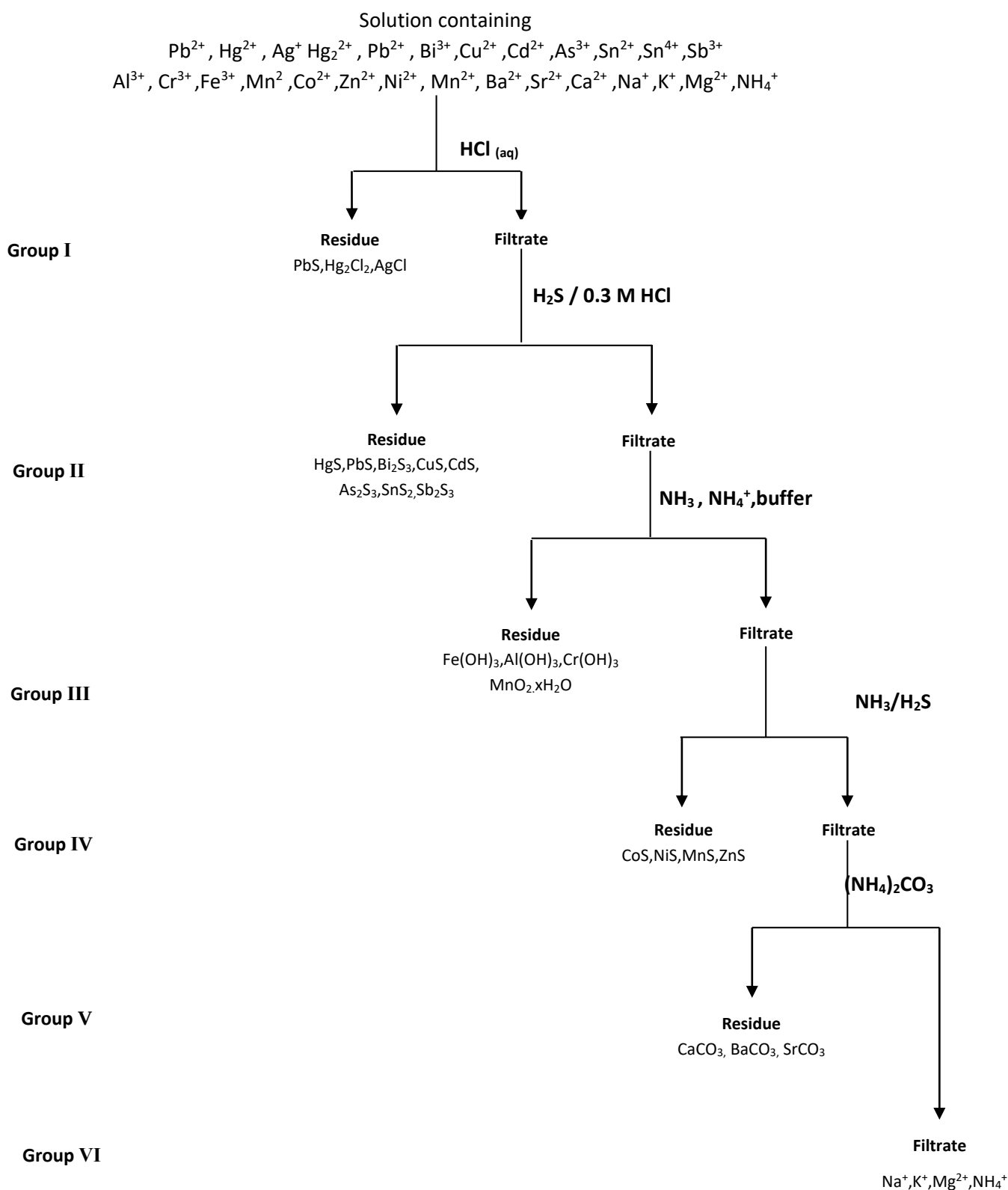
Tests for cations

First it is necessary to get the substance into a solution. Therefore use small quantities (5-10 g) of the powdered solid and examine the solubility in the solvents in the order given below.

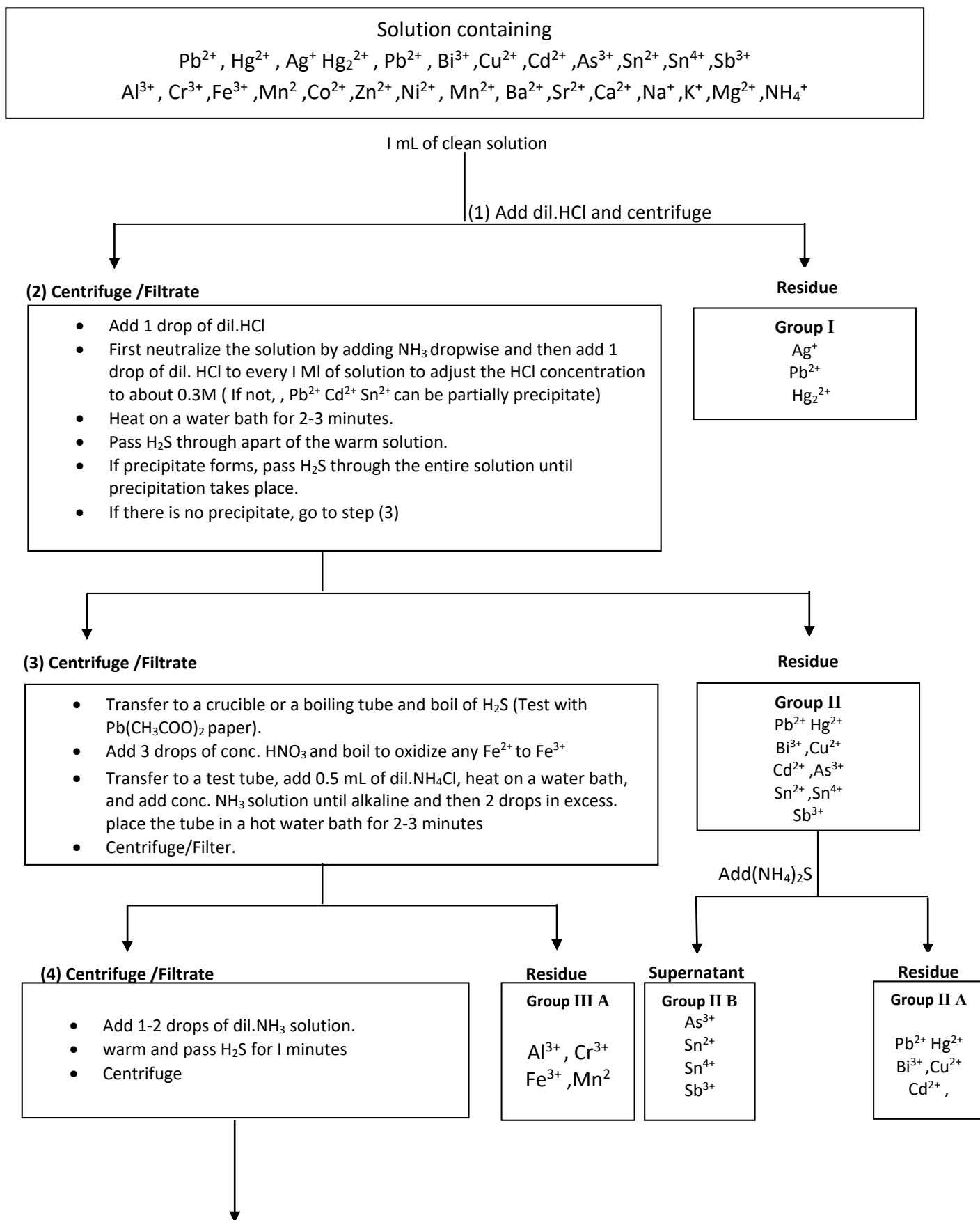
1. Distilled water (Cold/Hot)
 2. Dilute HCl
 3. Conc. HCl
 4. Dilute HNO_3
 5. Conc. HNO_3
 6. Aqua regia (3Conc. HCl: Conc: HNO_3)
- Try the solubility first in the cold and then warming.
 - If the compound is soluble in hot dilute HCl, cool it and filter off, if a precipitate (of Group I) forms.
 - If concentrated HCl is employed for solution, dilute it and separate if Group I present and evaporate most of the acid.

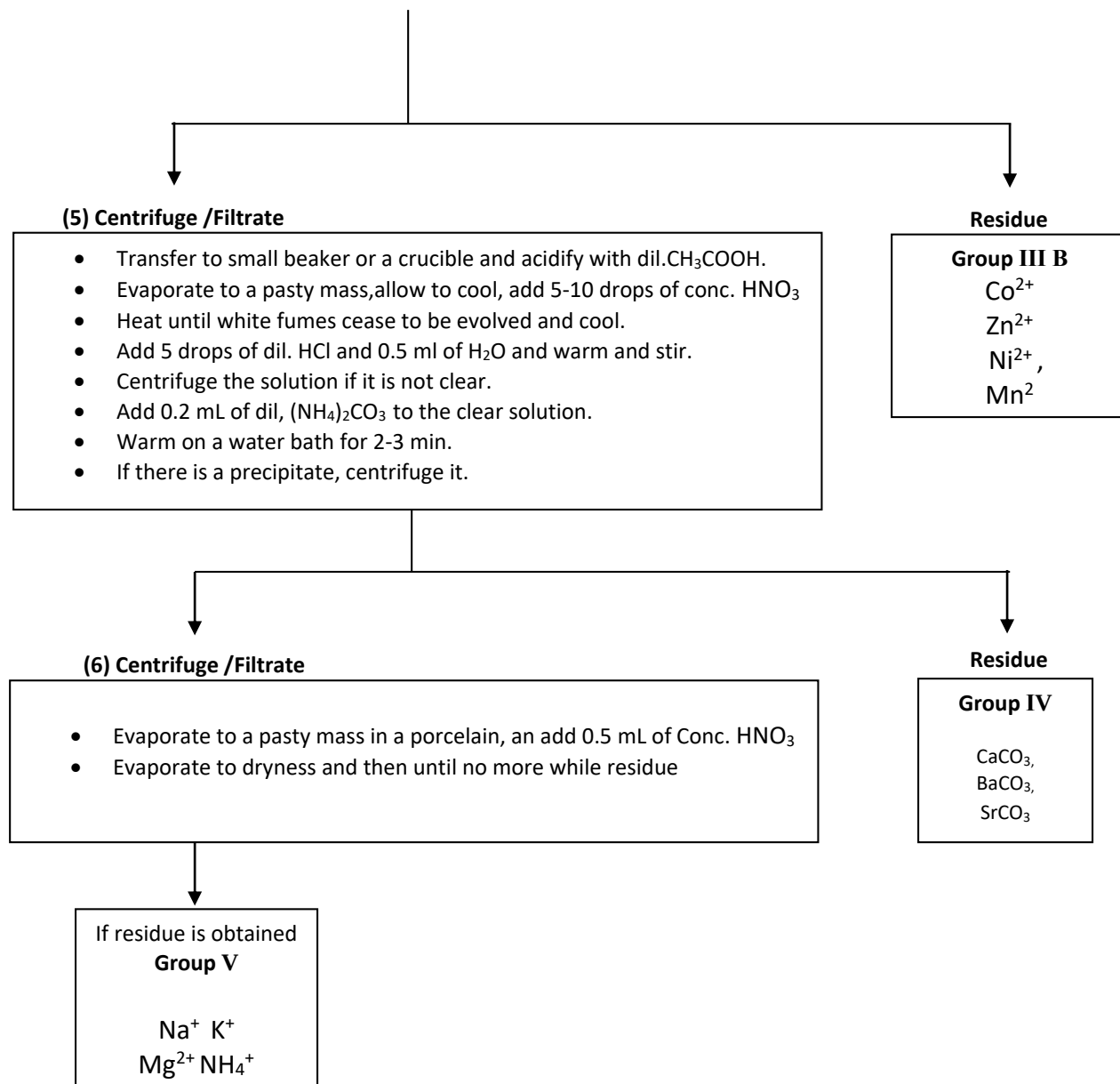
- When HNO_3 has been used for the process of solution, all the acid must be removed by evaporating nearly to dryness, adding a little HCl , evaporating again to a small bulk and then diluting with water.

Separation of cations into Groups



Separation of cations in to groups: Detailed procedure

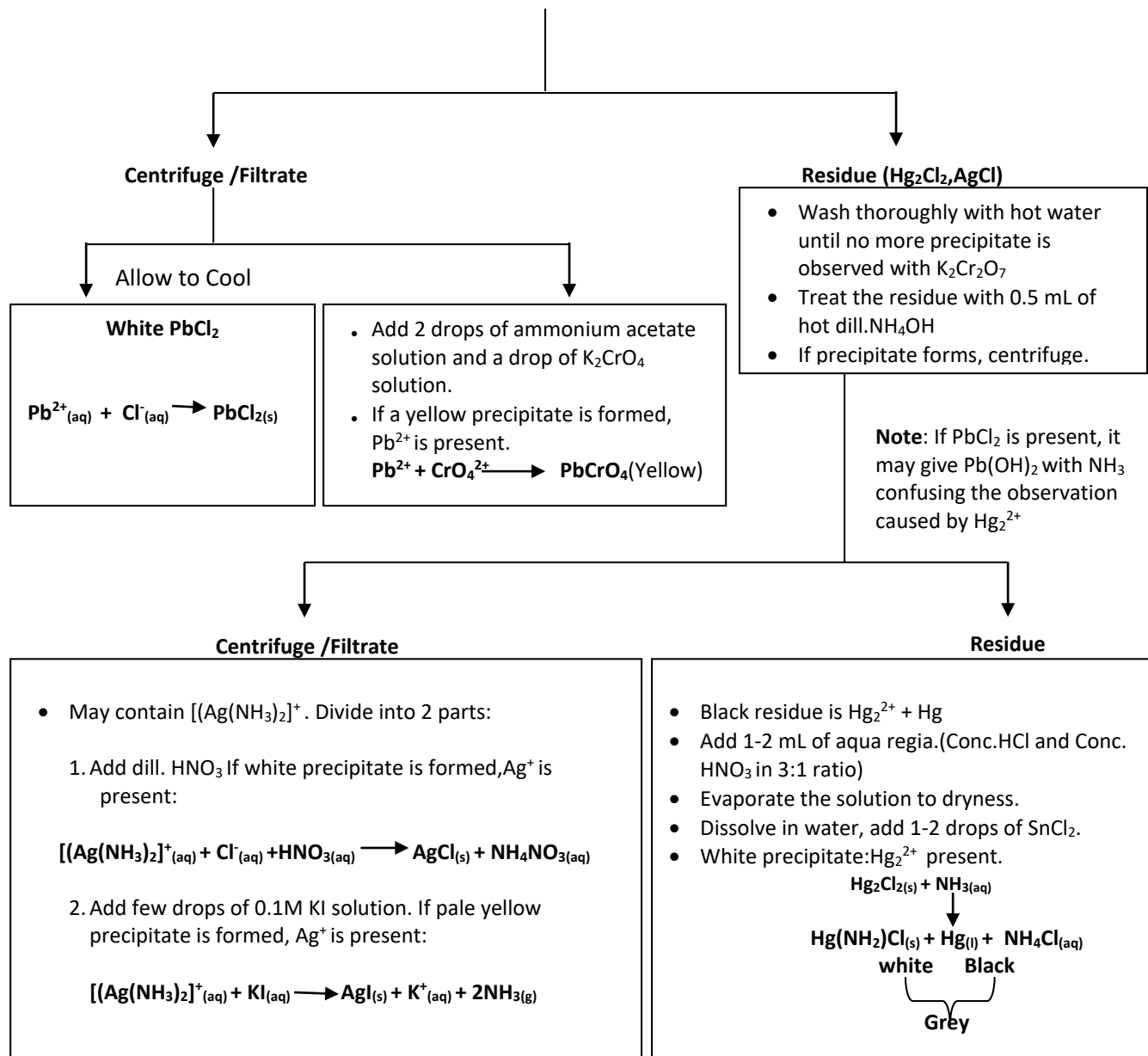




Group Separation

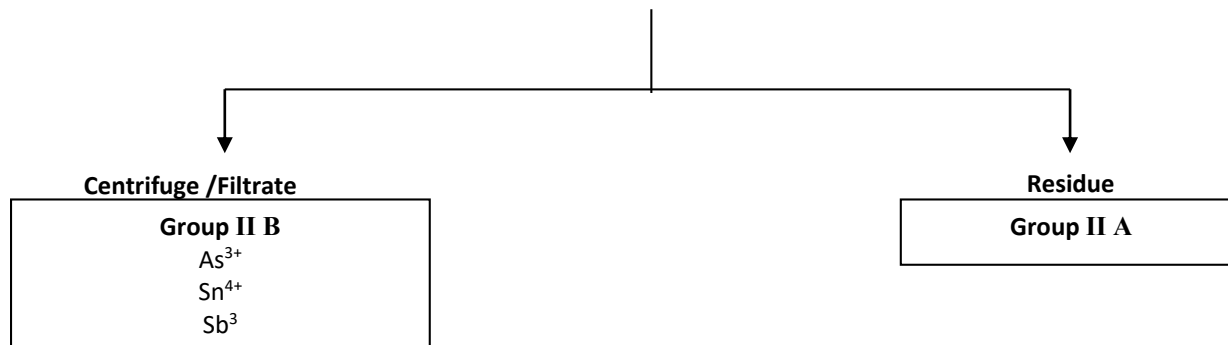
Separation within Group I

Wash the precipitate with 2-3 drops of distilled water. Add 1 mL of hot water to the precipitate. Place the tube in a boiling water bath for 1-2 minutes. If there is a precipitate, centrifuge and separate the solution from the residue.



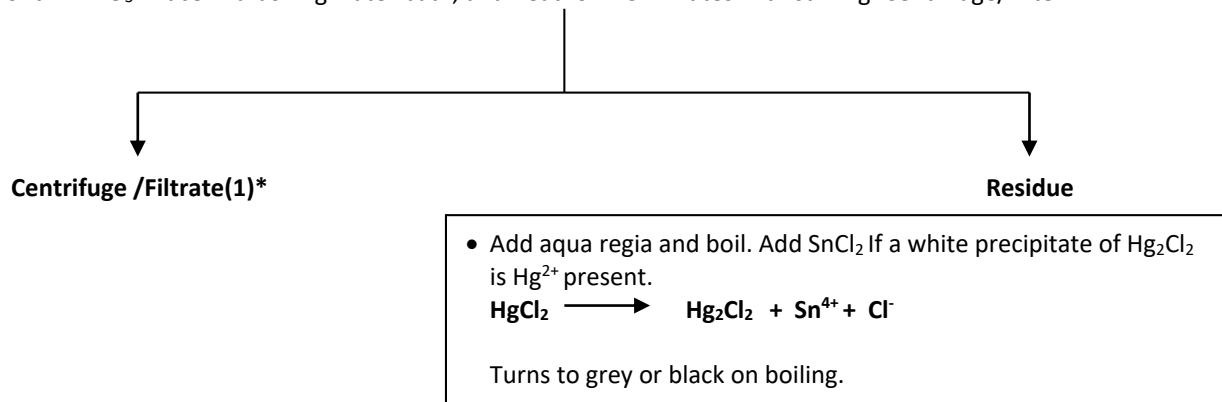
Separation of Group II

Wash the precipitate with small amount of NH_4Cl solution saturated with H_2S . Add few drops of $(\text{NH}_4)_2\text{S}_x$ and centrifuge the precipitate



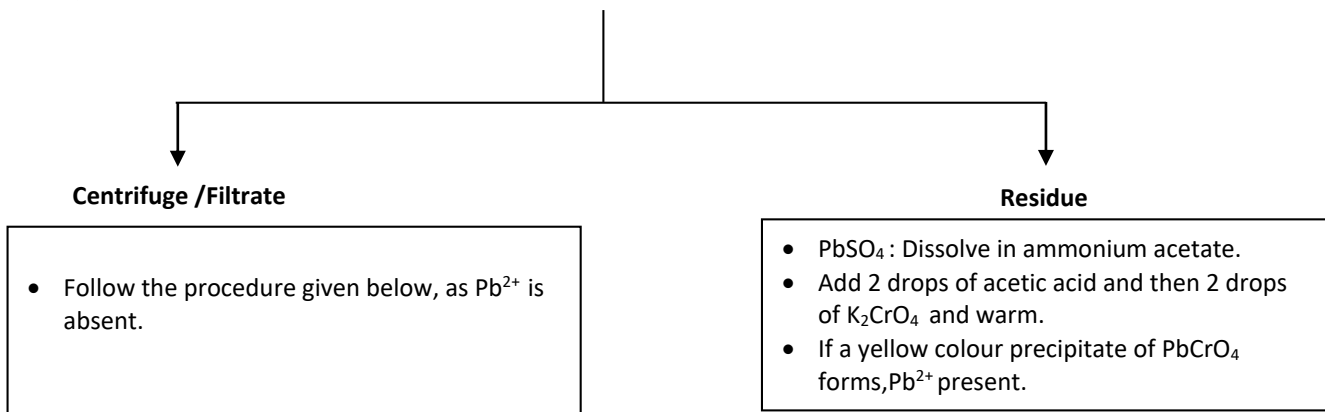
Separation of Group II A ($\text{HgS}, \text{PbS}, \text{Bi}_2\text{S}_3, \text{CuS}, \text{CdS},$)

Wash the residue once with 0.5 ml water and combine the washings with first centrifugate. Treat the precipitate with 1-1.5 mL of dil. HNO_3 . Place in a boiling water bath, and heat for 2-3 minutes with stirring. Centrifuge/Filter.



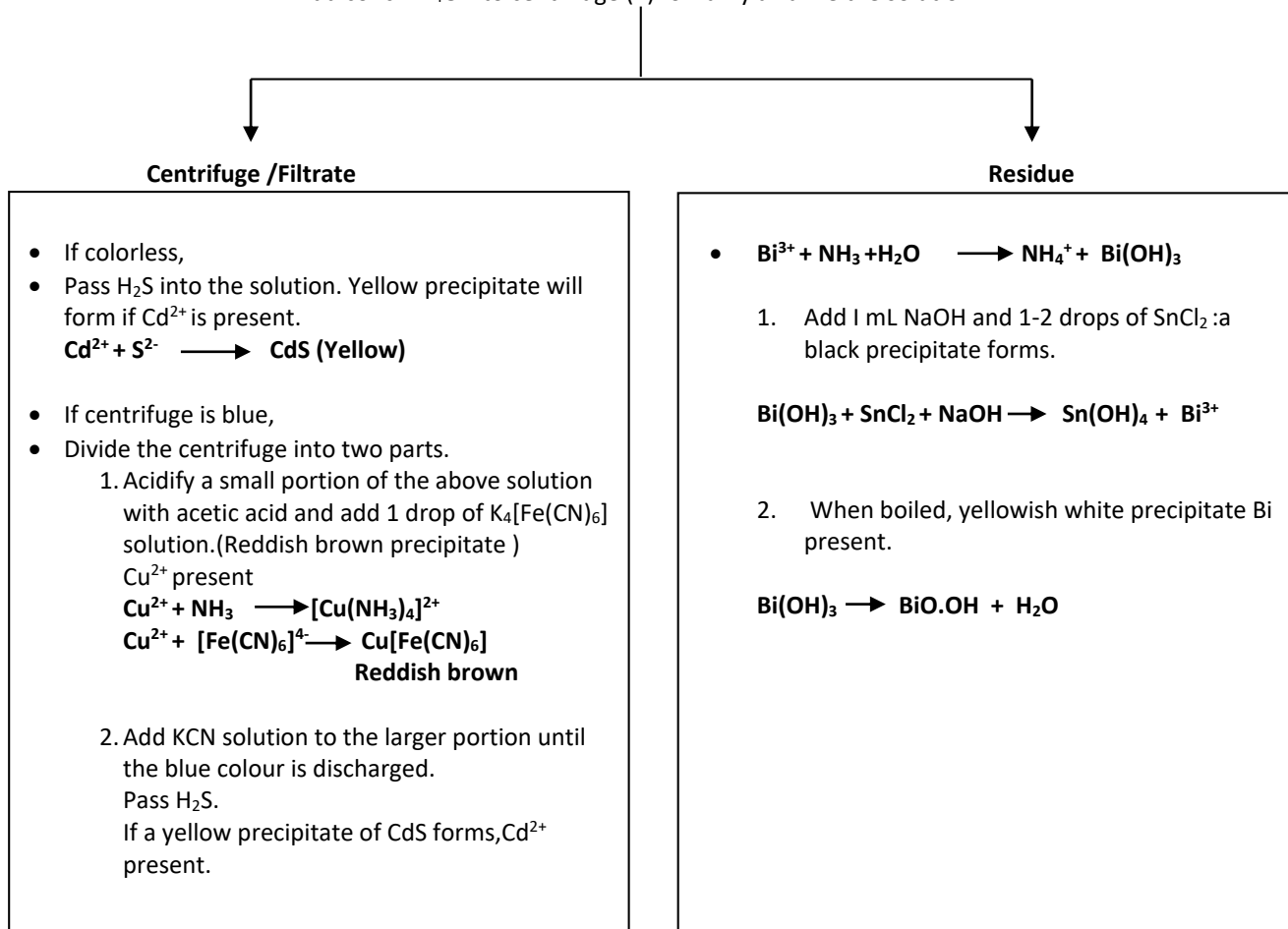
Analysis of centrifugate (1)* of group IIA. If Pb^{2+} present in Group I

Treat the centrifuge/Filtrate (1)* with dil. H_2SO_4 and add ethanol. If a precipitate forms, continue the addition until no more precipitate forms.



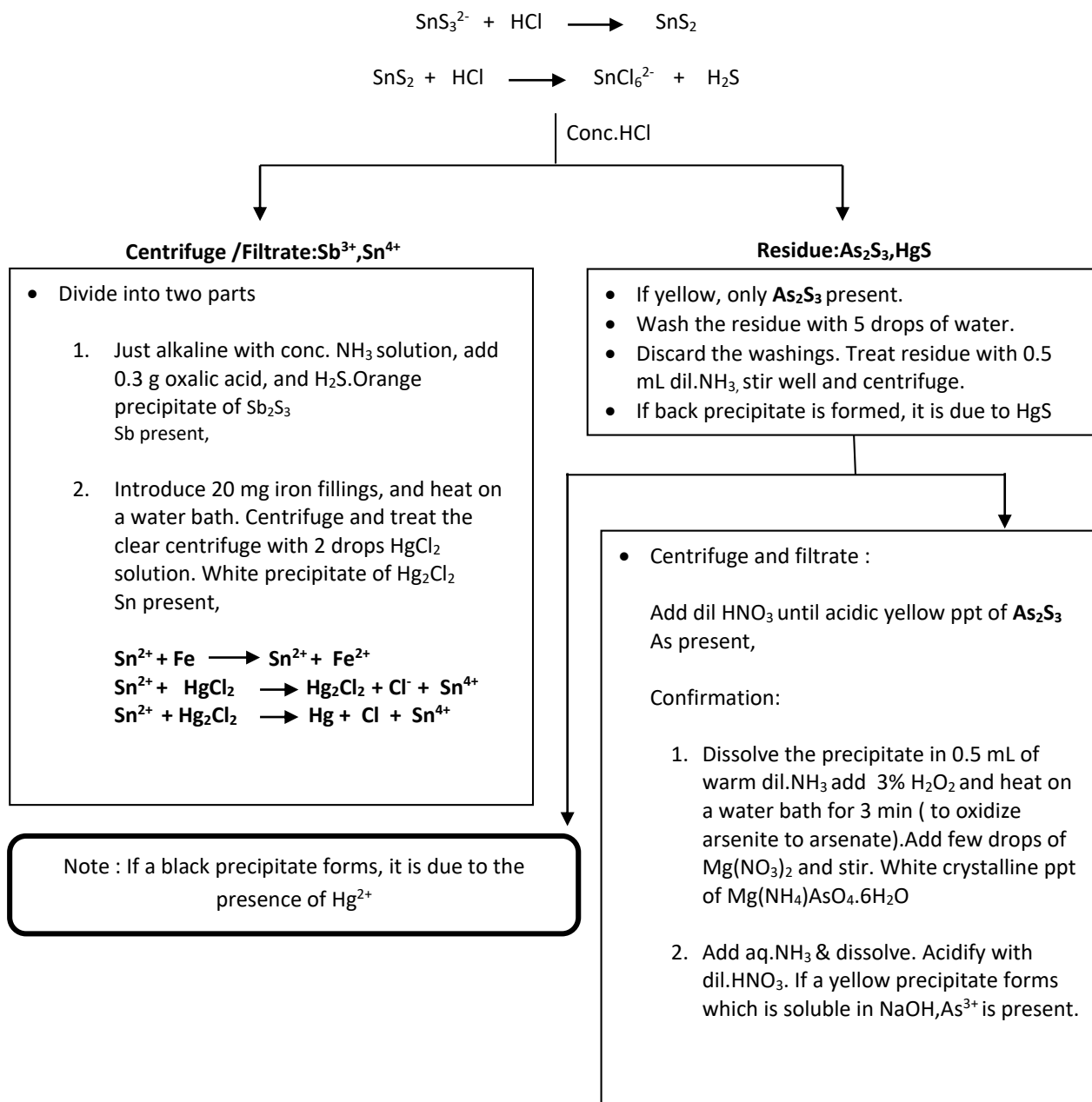
Analysis of centrifugate/filtrate (1)* of group IIA If Pb^{2+} absent

Add conc NH_4OH to centrifuge (1) for fairly alkaline the solution.



Separation of Group II B

Transfer the centrifuge to a small conical flask. Add conc. HCl drop wise with stirring until the mixture is distinctly acid to litmus paper, warm and centrifuge. A yellow precipitate indicates: As_2S_3 , SnS_2 , and Sb_2S_3



Removal of interfering ions (phosphate) from the solutions before the precipitation of Group III cations

- If PO_4^{3-} has been found removal of PO_4^{3-} is essential before performing group III. Dissolve the precipitate produced by the action of NH_4Cl and slight excess of NH_3 solution in a minimum volume of 2M HCl . Test for any Fe^{2+} with $\text{K}_4[\text{Fe}(\text{CN})_6]$ or NH_4SCN .
- Cool the solution.
- Add 2M NH_3 drop wise, stirring until either a faint permanent precipitate is obtained or until the solution is just alkaline.
- Add a few drops of dil. Acetic acid and a few drops of ammonium acetate solution.
- Add neutral FeCl_3 solution drop wise until the solution acquires a deep brownish red colour. (If the solution is already red in colour, omit this step.)
- Dilute the solution with hot water.

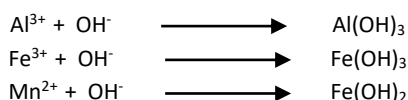
Residue: Group IIIA

Add H_2O , NaOH , and H_2O_2 and boil until O_2 is Evolved. Filter.

Centrifugate: Group IIIB

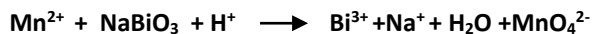
Separation of Group III A

Wash with little water and transfer the precipitate with aid of 10 drops of distilled water to a crucible, add 1-2mL of NaOH and 2 drops 3% H_2O_2 solution. Boil gently until the evolution of O_2 ceases (about 1 minute). Transfer the mixture to a centrifuge tube and Centrifuge.



Residue: $\text{MnO}(\text{OH})_2, \text{Fe}(\text{OH})_3$

- Wash with a little hot water.
- Dissolve the precipitate in 0.5 mL dil. HNO_3 , if necessary, 2 drops of 3% H_2O_2 or saturated $\text{H}_2\text{S}_2\text{O}_3$.
- Warm on a water bath.
- Divide the solution into 2 parts.
 1. Add 1 drop of $\text{K}_4[\text{Fe}(\text{CN})_6]$: Blue ppt. Fe^{3+} is present.
 2. Dilute with 1 mL of water, cool, add 10 mg NaBiO_3 shake and allow the solid to settle. Violet solution of MnO_4^{2-} indicates the Mn^{2+}



Centrifugate: CrO_4^{2-} (Yellow), $\text{Al}(\text{OH})_3$

- If yellow Cr^{3+} is present,
- Divide the solution into 2 parts.
- 1. Acidify with dilute acetic acid and add 1 drop of $\text{Pb}(\text{OAc})_2$.
Yellow precipitate of PbCrO_4 . Cr^{3+} present.
 2. Acidify with dilute acetic acid and add 1 drop of 1% "Aluminon" $(\text{NH}_4)_2\text{CO}_3$ solutions.

Add NH_4OH and $(\text{NH}_4)_2\text{CO}_3$ solutions.
A red precipitate indicates the presence of Al^{3+}

Cool and stir the precipitate with 1 mL of dil. HCl 1-2 minutes. Centrifuge

Centrifugate: Mn^{2+} , Zn^{2+}

- Transfer to a boiling tube/crucible, boil to expel H_2S . (Test with $\text{Pb}(\text{OAc})_2$ paper).
- Return liquid to ignition tube, cool, add excess NaOH solution.(0.5-1 mL) and 4 drops of 3% H_2O_2 solution.
- Heat on a water bath for 3 minutes. Centrifuge.

Centrifugate: $\text{Zn}(\text{OH})_4^{2-}$

- Divide into 2 parts
 1. Acidify with dil.HOAc and pass H_2S . White precipitate of ZnS : Zn^{2+} present
 2. Acidify with dill. H_2SO_4 , add a drops of dil. $\text{Co}(\text{CH}_3\text{COO})_2$ or cobalt nitrate, 0.5 mL NH_3 of ammonium tetrathiocyanato mercurate, stir. Pale-blue precipitate: Zn present.

Residue : $\text{MnO}(\text{OH})_2$

- Dissolve the ppt.in 0.5 mL dil HNO_3 and a drop 3% H_2O_2 solution.
- Warm on a water bath for 2 min to decompose excess of H_2O_2 cool.
- Add 50 mg NaBiO_3 shake and allow to settle. Purple solution of MnO_4^- : Mn^{2+} present.

Residue: CoS , NiS

- If black, may contain CoS and NiS . Test residue with borax bead test. Blue bead: Co present.
- Add either 10-15 drops of dil HCl and 5 drops of dissolve the precipitate in a mixture of NaOCl solution and HCl.
- Add dil. HCl boil until Cl_2 is expelled. Cool and dilute.

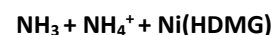
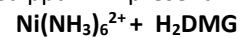
Divide the solution into 2 parts.

1. Add 0.5-1 mL amyl alcohol and 50 mg of solid NH_4SCN and shake.

Blue coloration in the alcohol layer Co^{2+} present.

2. Add 1 drop of NH_4Cl solution, render faintly alkaline with dill NH_3 solution, and add 3-5 drops of dimethyl glyoxime reagent.

Red ppt : Ni^{2+} present



Separation of Group IV

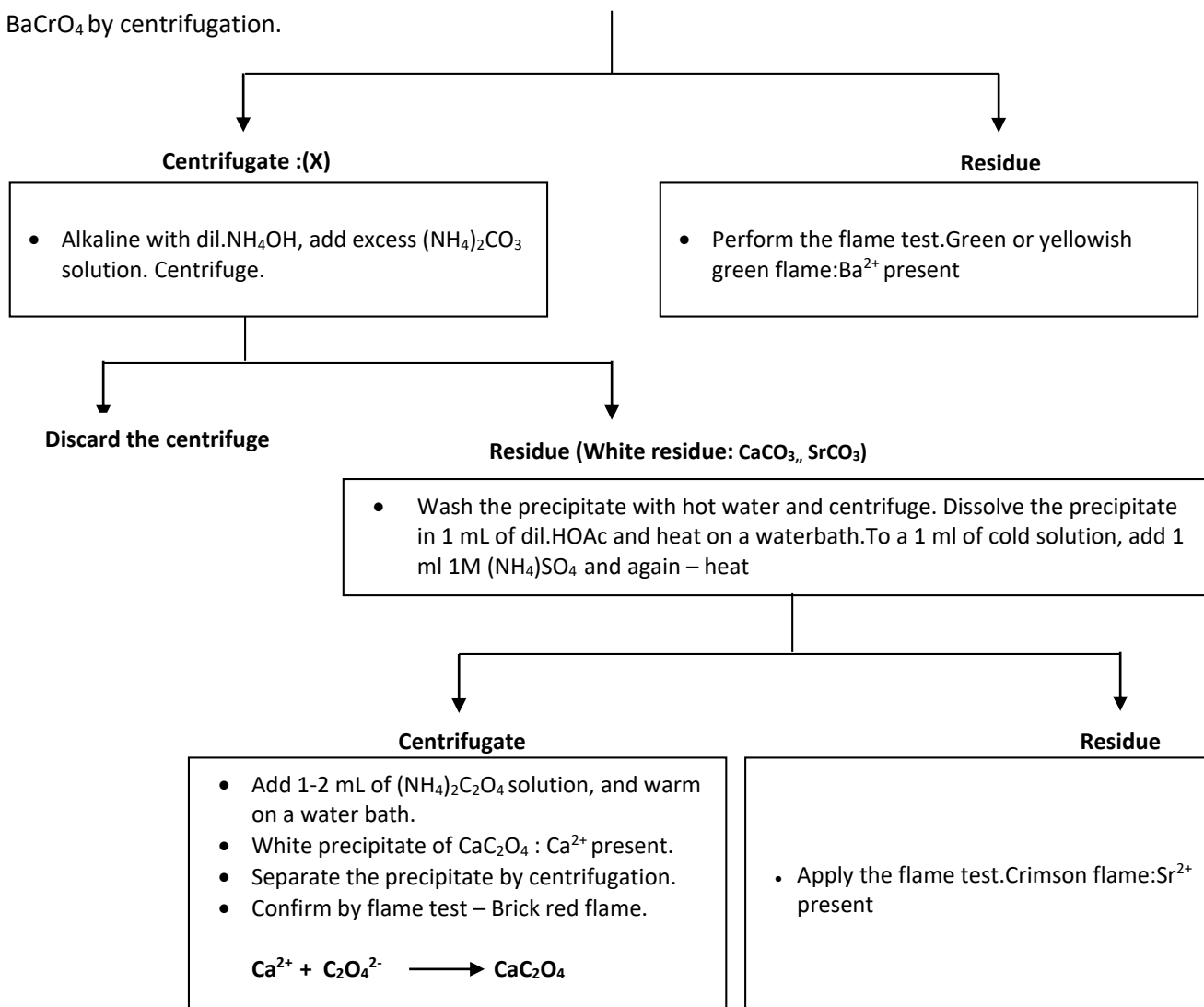
Wash the precipitate with hot distilled water. Treat the precipitate with 0.5 mL dilute HOAc and stir. Place in a hot water bath until the precipitate has dissolved. Dilute with 0.5 mL water. Test 3-4 drops of the hot solution for Ba^{2+} by adding a drop or two drops of $\text{K}_2\text{Cr}_2\text{O}_7$ solution. A yellow precipitate is obtained if Ba^{2+} is present.

If Ba^{2+} is absent,

Discard the portion used for Ba^{2+} , warm the remainder of the solution to expel CO_2 , centrifuge and follow step (X).

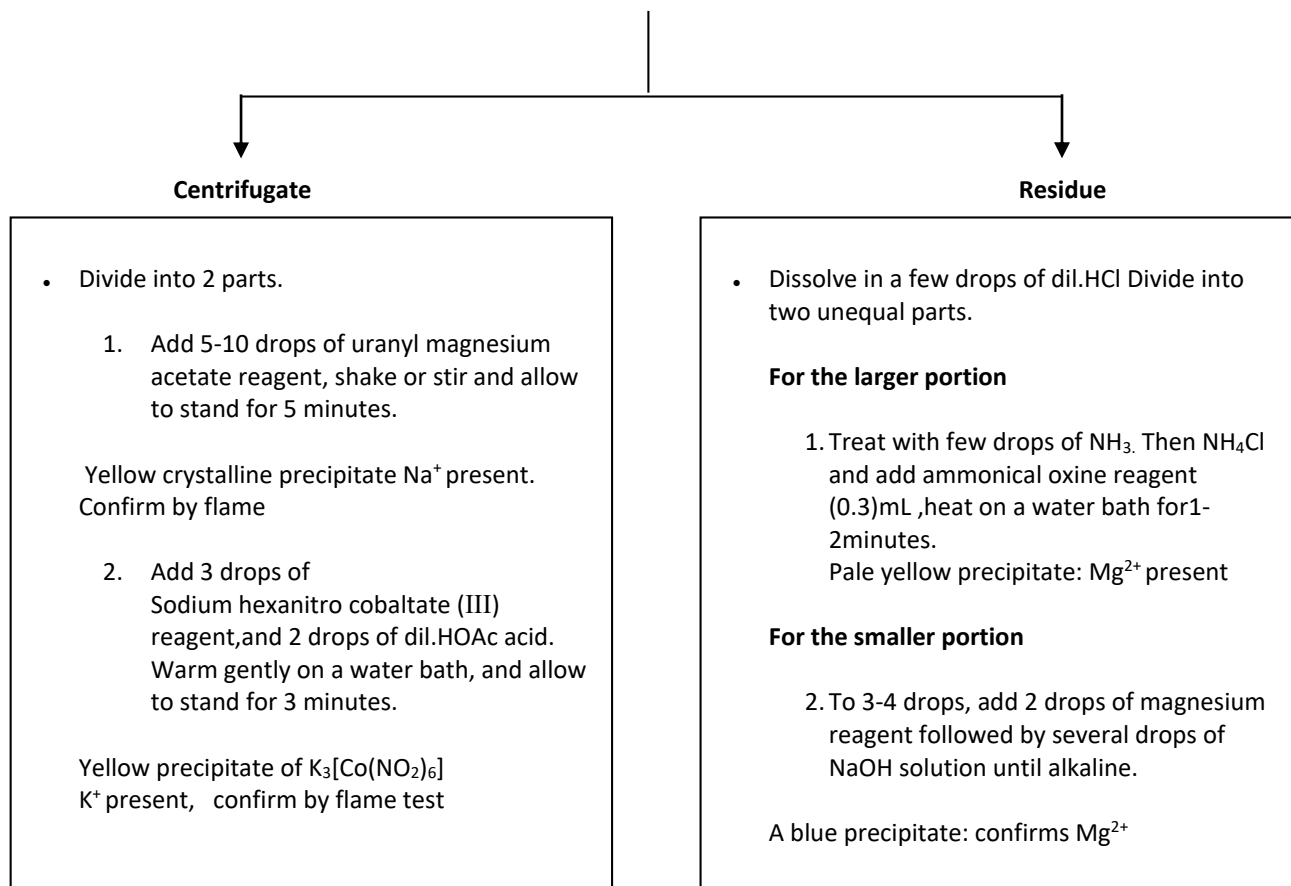
If Ba^{2+} present,

To the remainder of the hot solution add a slight excess of K_2CrO_4 solution, and separate the precipitate of BaCrO_4 by centrifugation.



Separation of Group V

Treat the dry residue with 1 mL water, stir and transfer with the aid of a further 0.5 mL water to centrifuge tube & centrifuge.



Reference : Vogel's Qualitative Analysis ; Seventh edition; Page numbers : 268 - 277